10.8.2026: Kangaroo rat frequency distributions over size, sex and seasons

Set up

```
library(ggplot2)
library(brms)
```

```
Loading required package: Rcpp
```

```
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':

    ar
```

```
library(ggeffects)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.4     v tibble    3.3.1
v purrr     1.2.0     v tidyr     1.3.2
```

```
-- Conflicts --------------------------------------- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(readxl)
library(here)
```

```
here() starts at C:/Users/Rwncr/Documents/OFFICIALKRATBUSINESS/Krat-Senior-thesis-2026-27
```

```
library(dplyr)
library(ggtext)

#Load in clean data set with all values
df <- read.csv("RC2026_KRAT_DATA.08.01.csv")
```

Goal

Create frequency distributions for general sizes of each species.

Create frequency distributions for each species over the different seasons of the year.

Our data demographics

36 heermanni, 24 venustus

```
#Remove rows with the most NAs in continuous, quantitative variables or where most measureme
df <- df[-c(6, 10, 11, 12,13, 14,15, 16, 19, 20, 21, 22, 23, 24, 25, 34, 36, 37, 38, 39, 51,

#Show number of each species included in analysis: 36 heermanni, 24 venustus
table(df$Species)
```

```
Dipodomys heermanni  Dipodomys venustus
                 36                  24
```

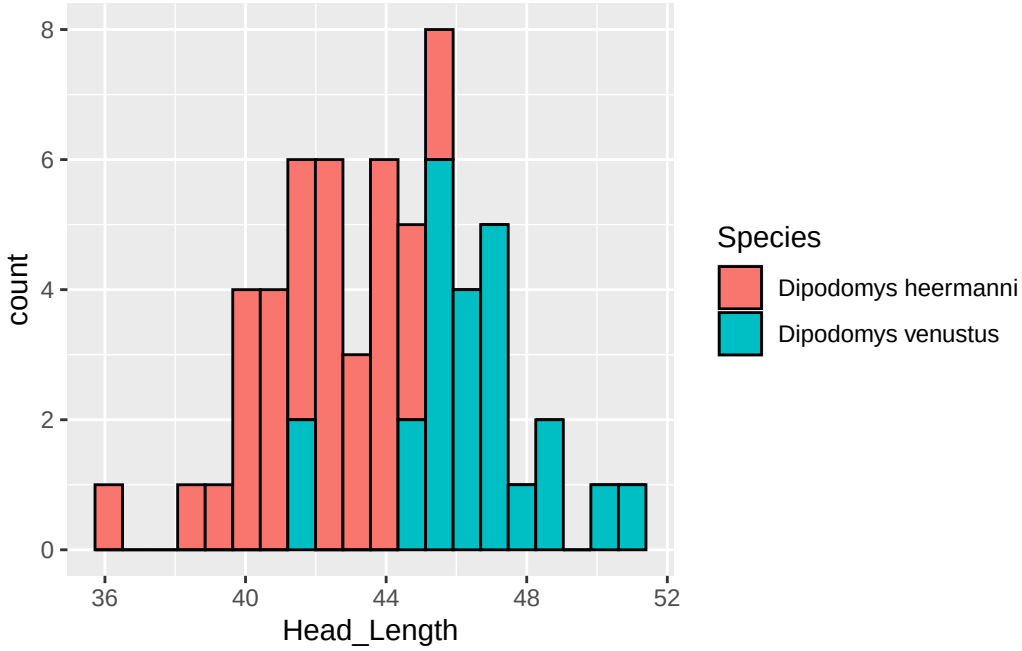
General size frequency distributions for Head Length

With all data points as of 8.01.2026

Option here between a histogram and a density distribution. I like the density one better for looks, but probably tells you less if you don't know about the data for counts and the actual frequency.

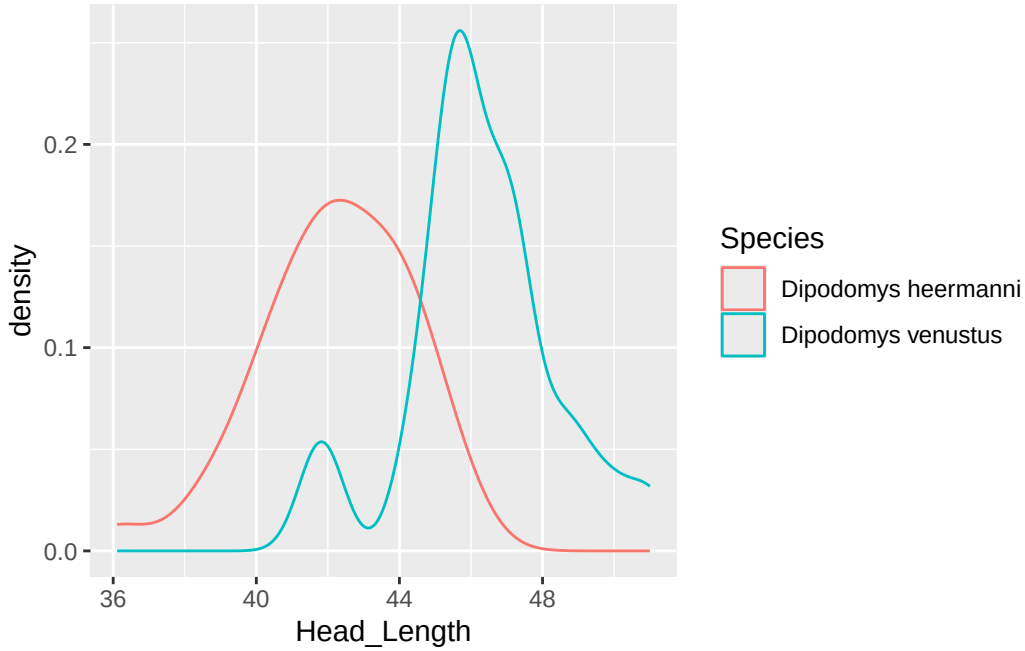
```
#Create a histogram of the head length, separating by species
df %>%
  ggplot(aes(x= Head_Length, fill =  Species))+
geom_histogram(bins = 20, color = "black")
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_bin()`).
```




```
#Create a density, smooth line frequency distribution, separating by species
df %>%
  ggplot(aes(x= Head_Length, color =  Species))+
geom_density()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```



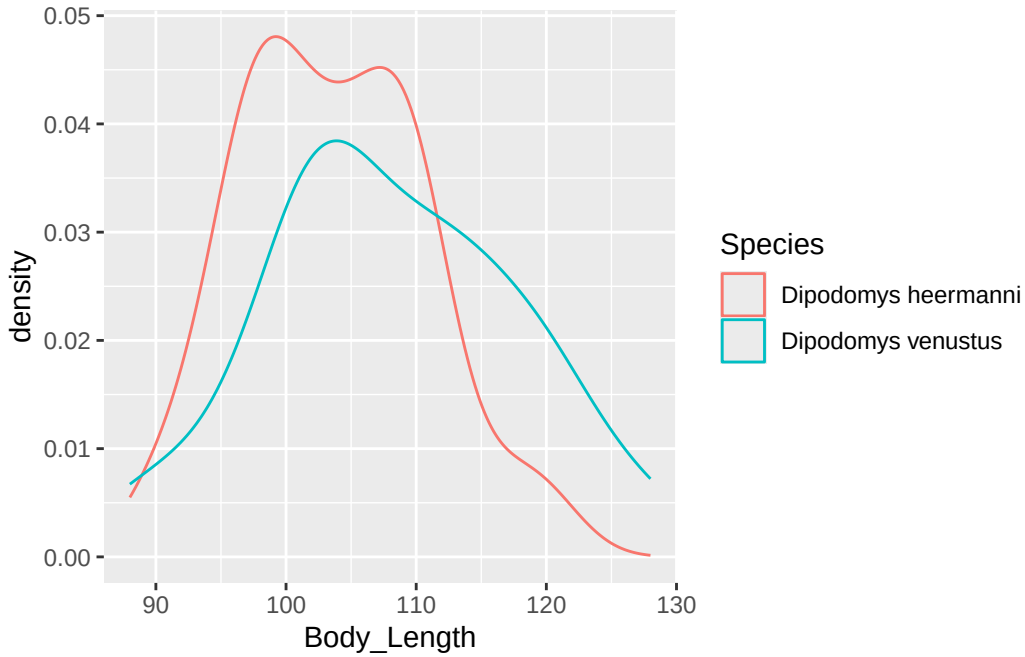
General size frequency distribution for Body Length

With all data points

This graph has the frequency distribution for body length of both species, showing they are not very different in body length, but in the past graphs the head length has differed a lot more. Maybe body length is a better indicator of general size despite it being more coarse??

```
#Create a density, smooth line frequency distribution, separating by species
df %>%
  ggplot(aes(x= Body_Length, color =  Species))+
geom_density()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```



Size over the seasons

Observations of both species per season:

Fall: 23, Spring: 3, Summer 33, Winter 1

Both species across the seasons

```
#make a new column for season
df$Season <- NA
 # it worked!

# Extract month and assign season
df <- df %>%
  mutate(
    Month = month(Date),
    Season = case_when(
      Month %in% c(12, 1, 2) ~ "Winter",
      Month %in% c(3, 4, 5)  ~ "Spring",
      Month %in% c(6, 7, 8)  ~ "Summer",
```

```
      Month %in% c(9, 10, 11) ~ "Fall"
    )
  )
# check that it worked
table(df$Season) #Fall 23, Spring 3, Summer 33, Winter 1
```

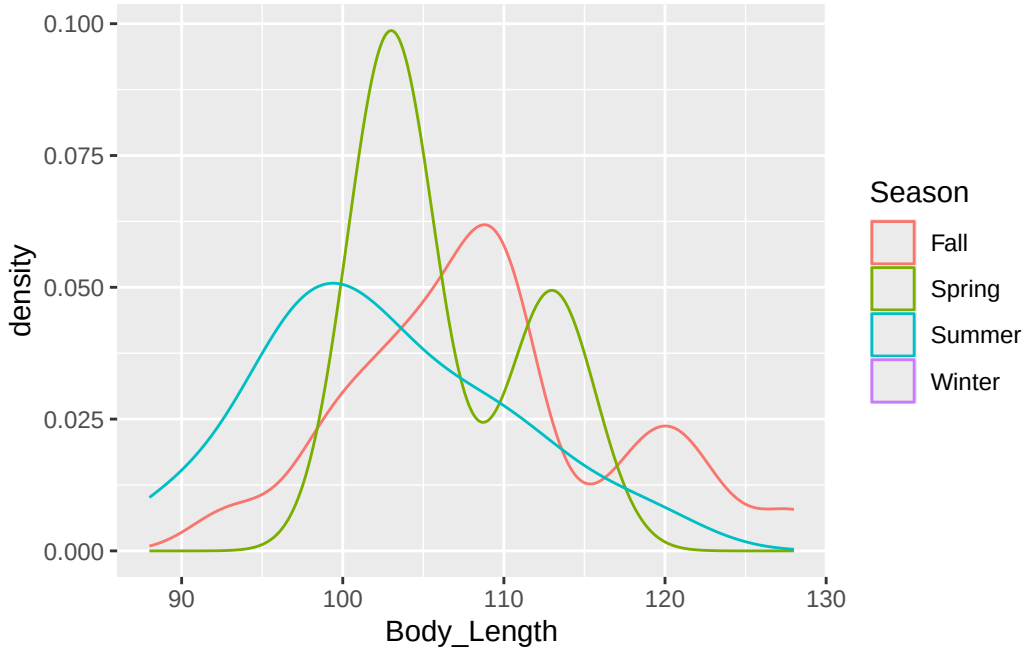
```
  Fall Spring Summer Winter
    23      3     33      1
```

```
#graph with color = season, of all kangaroo rats both species
#Create a density, smooth line frequency distribution, separating by season and including bot
df %>%
  ggplot(aes(x= Body_Length, color =Season))+
geom_density()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```

```
Warning: Groups with fewer than two data points have been dropped.
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_density()`).
```



Heermanni across the seasons

For just heermanni observations per season:

Fall: 14

Summer: 22

Winter: 0

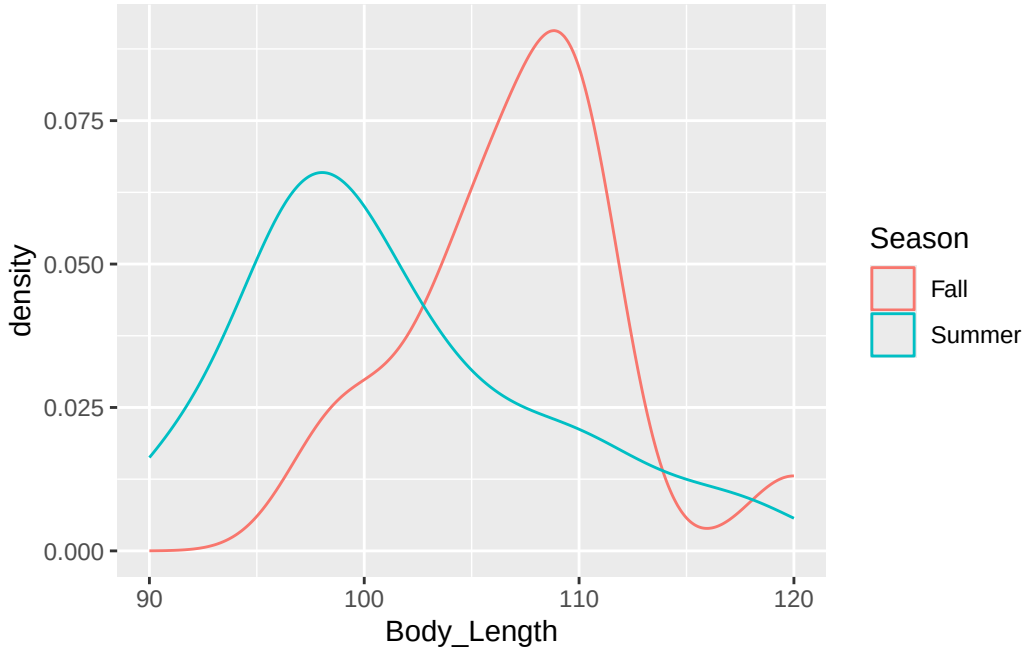
Spring: 0

```
#Filter for only D. heermanni
dfhe <- df |> filter(Species == "Dipodomys heermanni")

#Demographics 14 fall, 22 summer
table(dfhe$Season)
```

```
Fall Summer
  14     22
```

```
#Create a density, smooth line frequency distribution, separating by season
dfhe %>%
  ggplot(aes(x= Body_Length, color =Season))+
geom_density()
```



This seasonal graph of the heermanni is super interesting and might have enough of a sample size to be able to say something?!

Venustus over the seasons

For just venustus observations per season:

Fall: 9

Spring: 3

Summer: 11

Winter: 1

```
#Filter for only D. heermanni
dfve <- df |> filter(Species == "Dipodomys venustus")
```

```
#Demographics 14 fall, 22 summer
table(dfve$Season)
```

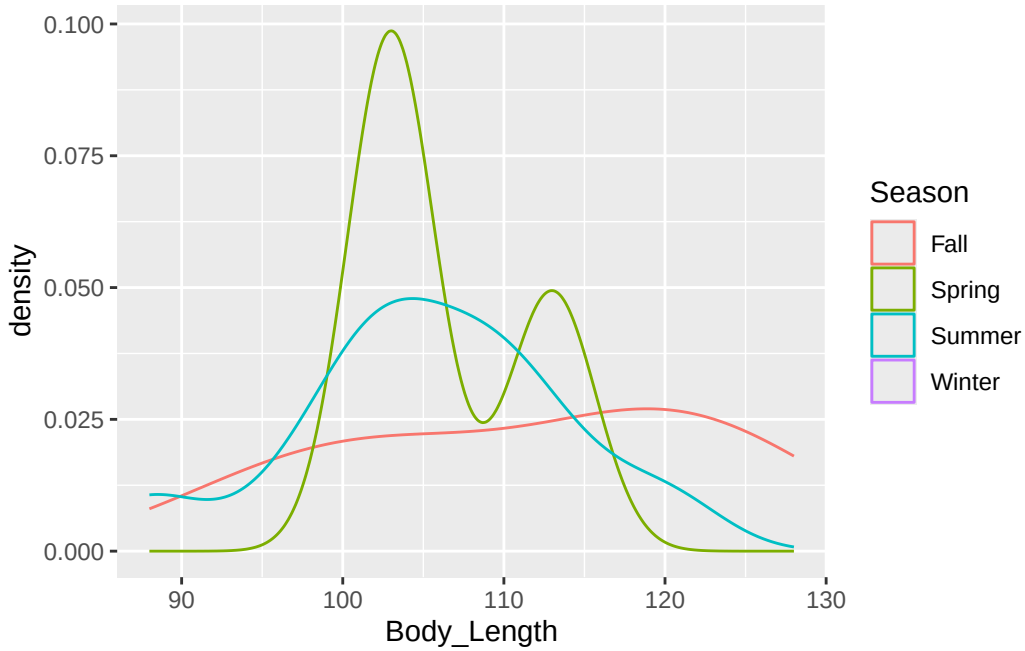
```
  Fall Spring Summer Winter
     9      3     11      1
```

```
#Create a density, smooth line frequency distribution, separating by season
dfve %>%
  ggplot(aes(x= Body_Length, color =Season))+
geom_density()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```

```
Warning: Groups with fewer than two data points have been dropped.
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_density()`).
```



This graph here is not based on enough data to mean much, but we don't see the same split in size over the seasons as with the heermanni. This could be due mostly to the low sample size.

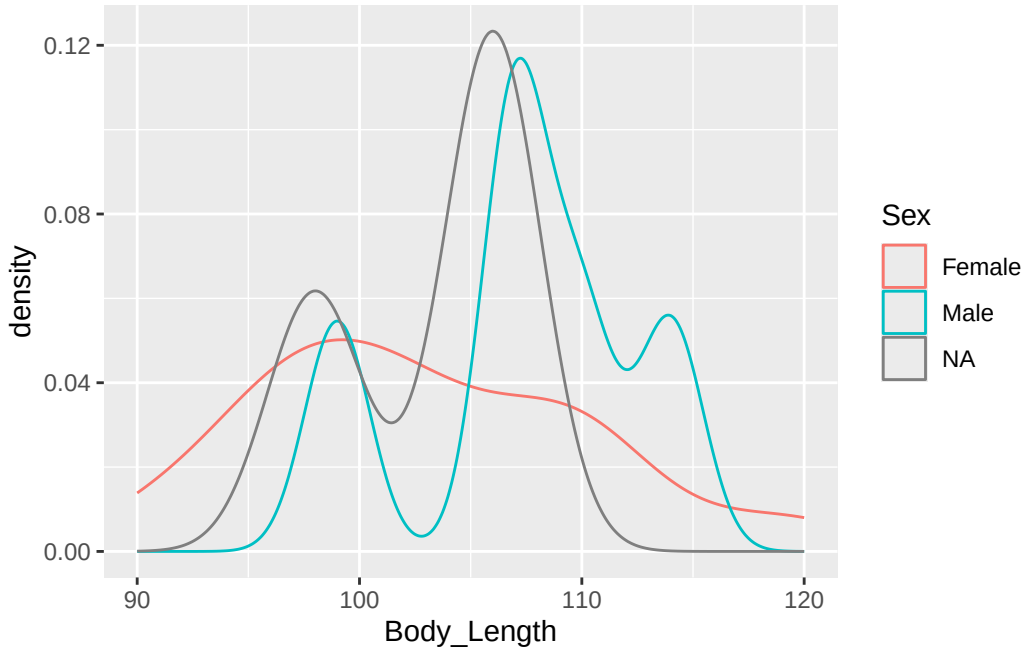
Seasonal sexual dimorphism?

Dipodomys heermanni

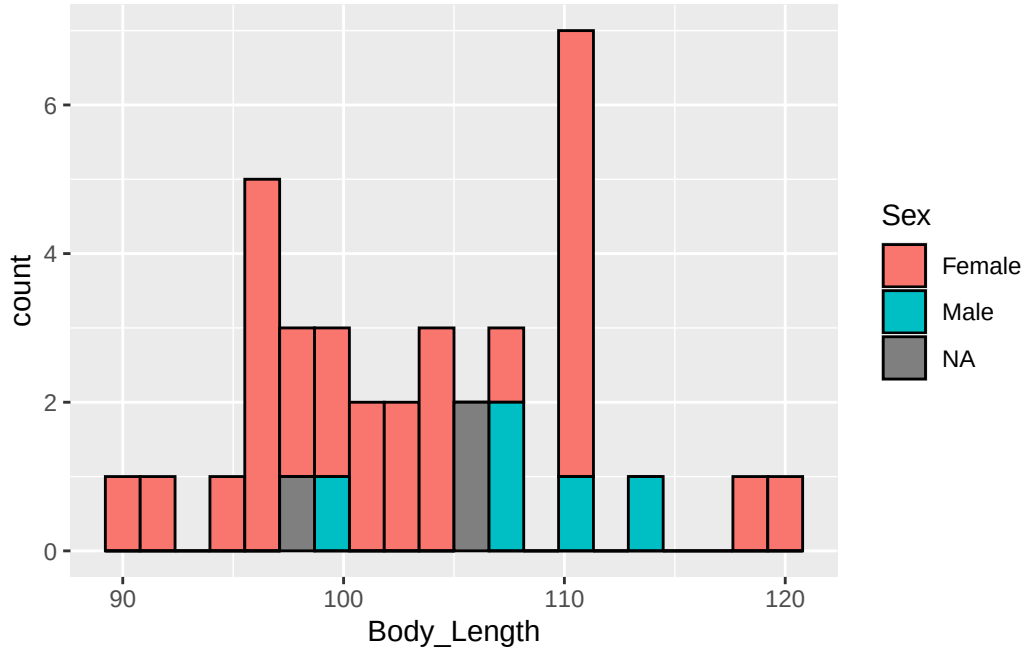
Observed 28 Female and 5 Male heermanni.

Females seem to be more abundant then males in both species or maybe they are out foraging more often due to greater metabolic needs (Lactating, pregnant, etc) and therefore get caught in the traps more often??

```
#Create a density, smooth line frequency distribution, of just the heermanni separating by se
dfhe %>%
  ggplot(aes(x= Body_Length, color =Sex))+
geom_density()
```




```
#Create a histogram of the body length, separating by sex of just the heermanni
dfhe %>%
  ggplot(aes(x= Body_Length, fill =  Sex))+
geom_histogram(bins = 20, color = "black")
```

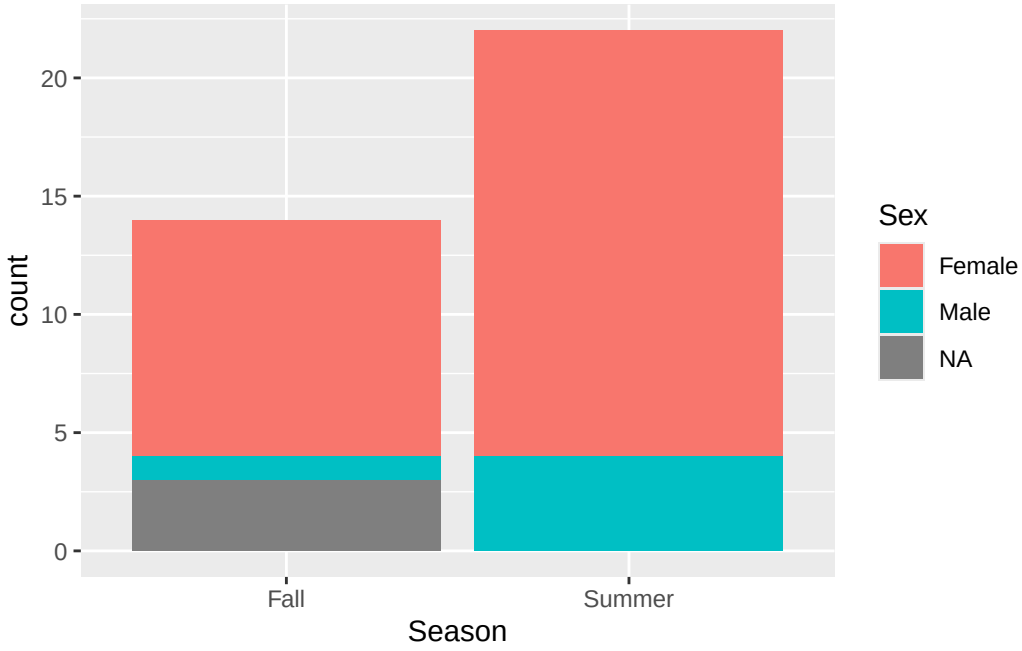



```
#Sexual demographics
table(dfhe$Sex)
```

```
Female   Male
    28      5
```

This map probably doesn't mean much due to the low sample size of males.

```
#Create a bar plot, separating by season and sex
dfhe %>%
  ggplot(aes(x= Season, fill=Sex))+
geom_bar()
```



There's way more females than males, I'm not sure we can say much from this.

Dipodomys venustus

Of the venustus observations:

F: 19, M: 1, Unknown: 1

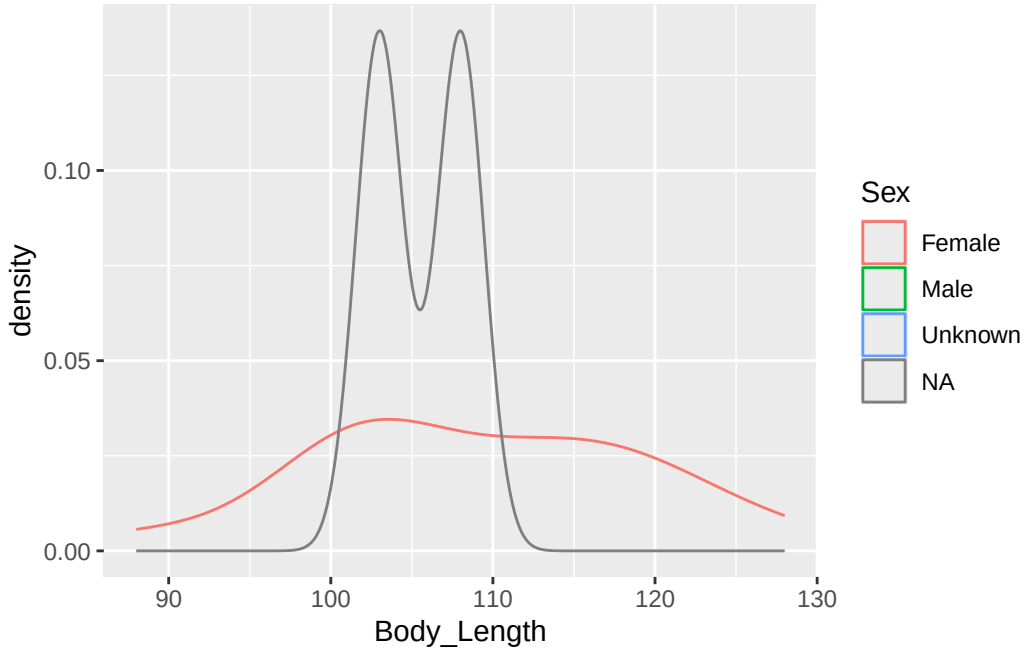
Graph doesn't even work because we only caught 1 male venustus at Hastings.

```
#Create a density, smooth line frequency distribution, separating by sex of the venustus and
dfve %>%
  ggplot(aes(x= Body_Length, color =Sex))+
geom_density()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```

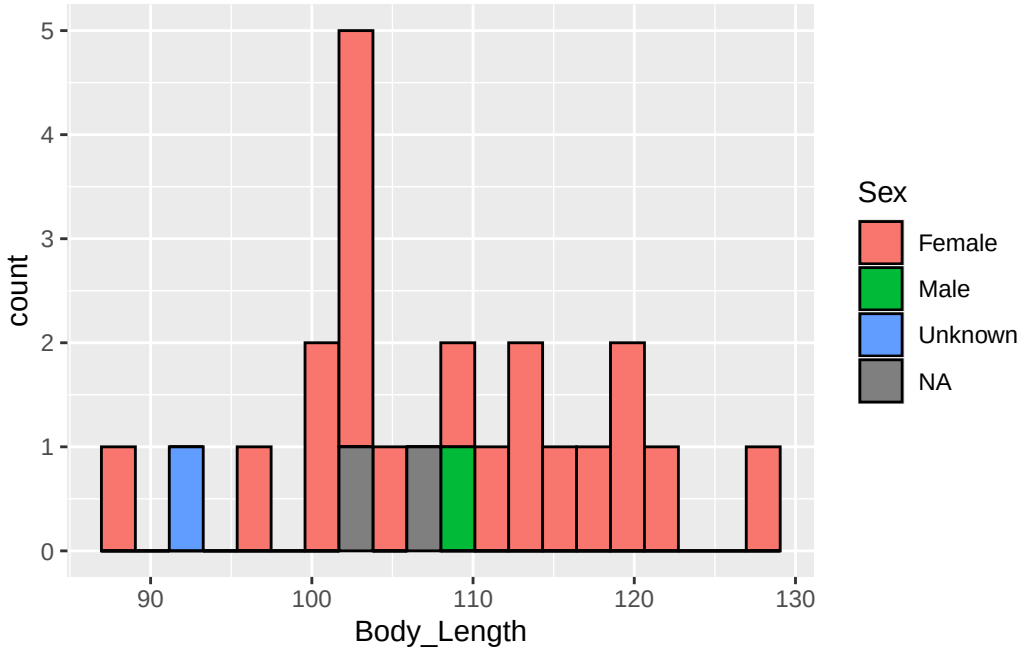
```
Warning: Groups with fewer than two data points have been dropped.
Groups with fewer than two data points have been dropped.
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_density()`).
```




```
#Create a histogram of the head length, separating by sex of just the venustus
dfve %>%
  ggplot(aes(x= Body_Length, fill =  Sex))+
geom_histogram(bins = 20, color = "black")
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_bin()`).
```




```
#Sexual demographics
table(dfve$Sex)
```

```
Female    Male Unknown
    19       1       1
```